C15 Alcohols – Knowledge Organiser

AQA A-level Chemistry • Set 1: Production and Oxidation **Paper 2**

#	Question	Answer
1	Equation for fermentation of glucose	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
2	Conditions for fermentation	Yeast, about 35 °C, anaerobic
3	Catalyst for hydration of ethene	H_3PO_4
4	Biofuel	Fuel made from biological material
5	Why bioethanol is claimed to be carbon neutral	CO_2 released = CO_2 absorbed by the crop
6	Oxidising agent for alcohols	Acidified potassium dichromate(VI)
7	Colour change of acidified dichromate	Orange to green
8	Oxidation products of a primary alcohol	Aldehyde, then carboxylic acid
9	Oxidation product of a secondary alcohol	Ketone
10	Tertiary alcohol with acidified dichromate	No reaction

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AQA A-level Chemistry • Set 2: Conditions, Products and Dehydration **Paper 2**

#	Question	Answer
1	Condition to obtain an aldehyde	Distil off as it forms
2	Condition to obtain a carboxylic acid	Heat under reflux, excess oxidising agent
3	Symbol for the oxidising agent in equations	[O]
4	Product of ethanol oxidised with distillation	Ethanal
5	Product of ethanol oxidised under reflux	Ethanoic acid
6	Oxidation product of propan-2-ol	Propanone
7	Catalyst for dehydration of alcohols	Conc H_2SO_4 or H_3PO_4
8	Products of dehydration of an alcohol	Alkene + water
9	Number of alkenes from dehydrating butan-2-ol	3
10	Benefit of making alkenes from alcohols	Polymers without using crude oil

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AQA A-level Chemistry • Set 3: Tests **Paper 2**

#	Question	Answer
1	Tollens' reagent	$[Ag(NH_3)_2]^+$
2	Positive result with Tollens'	Silver mirror
3	Positive result with Fehling's	Blue solution to brick-red precipitate
4	Active ion in Fehling's solution	Cu^{2+}
5	Ketones with Tollens' or Fehling's	No change
6	Test for a carboxylic acid	$NaHCO_3$
7	Positive result for a carboxylic acid	Effervescence
8	Gas given off by a carboxylic acid with $NaHCO_3$	CO_2
9	Primary alcohol	–OH carbon bonded to one other carbon
10	Why reflux is used	Heats without losing volatile substances